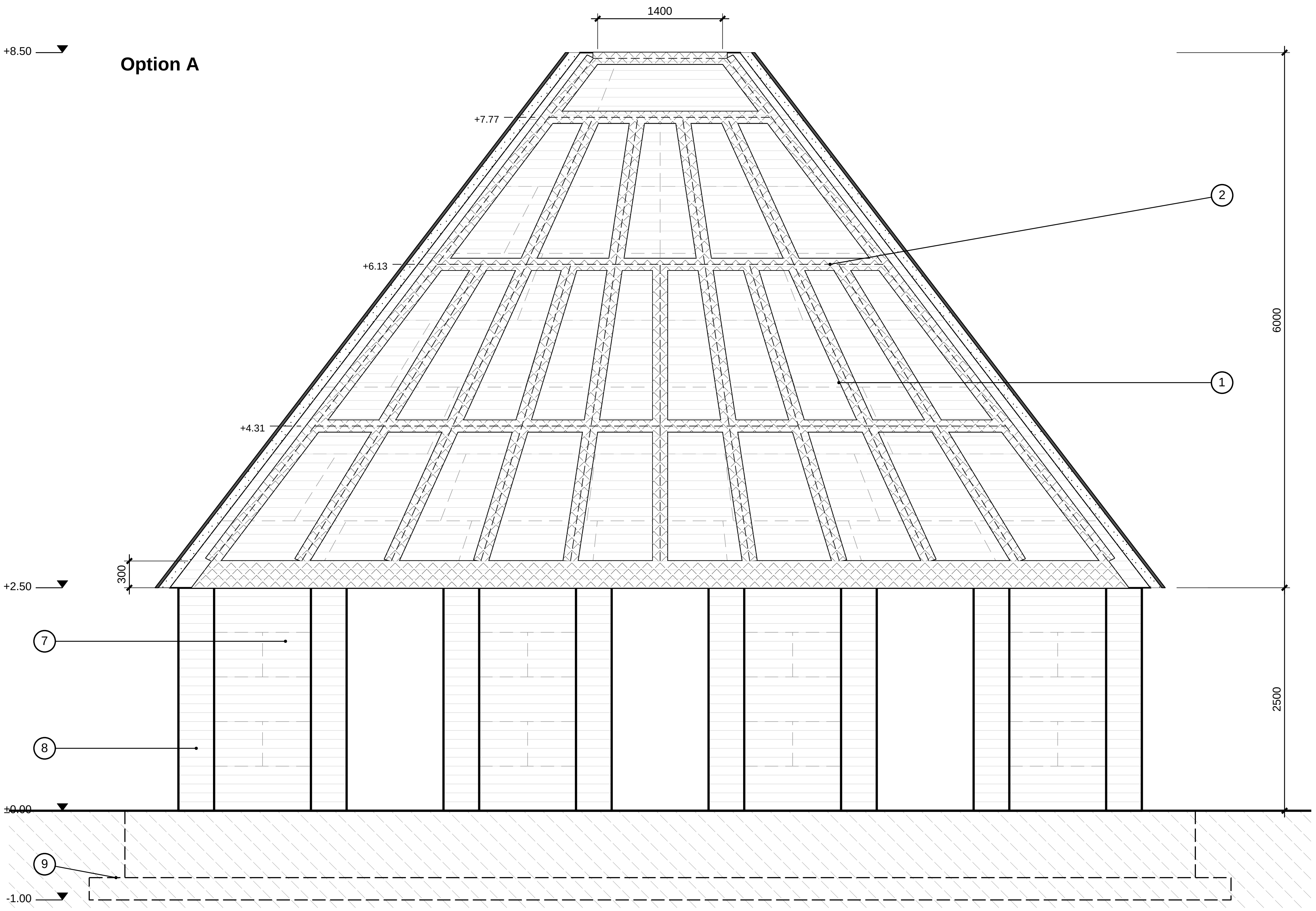
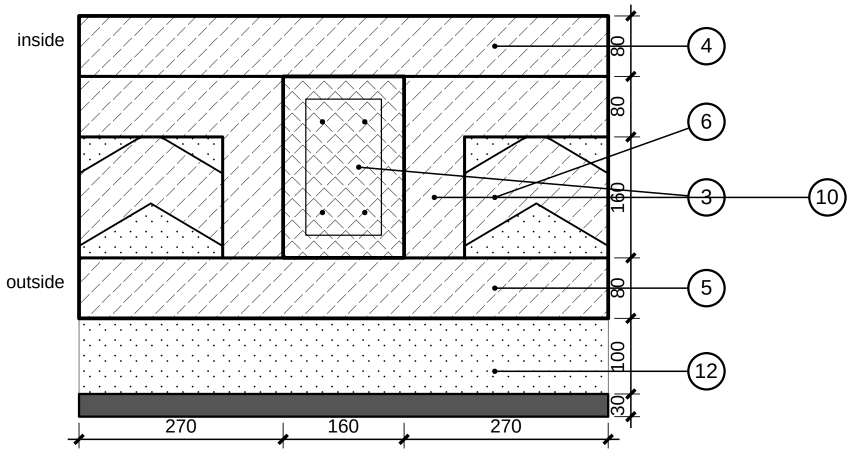
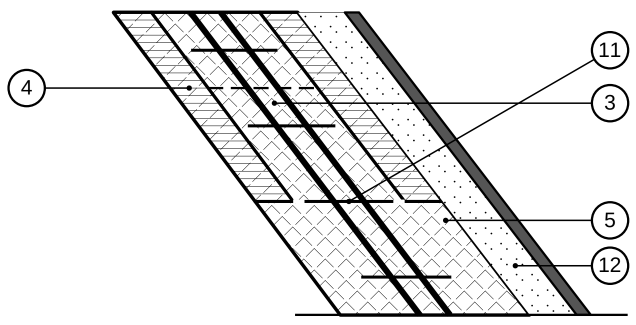


Plan cut through a rib at one layer, 1:10



Section along a rib at the eave, 1:20



Roof grid A: ruled-surface columns

one roof face seen square on

(1) Columns run in straight lines from the base edge (every 1050) towards the crown edge, so they converge like the ruled surface of the sketch.

(2) When the spacing drops to 500 (about +6.13) every other column stops; at about +7.77 only the four hip columns go on to the crown ring.

(3*) Rib (column): a channel 160 wide x 240 deep, printed in the 160 air zone + layer 2 stays in place as formwork and is filled with cast concrete, 4 Ø16 + Ø8 stirrups every 200 (assumed). Fibre concrete could replace the bars (see notes).

(4) Inner layer 1, 80: continuous, covers the grid on the room side.

(5) Outer skin 80: continuous, covers the grid on the outside (cladding on it).

(6*) Between the ribs the air zone stays open for thermal insulation (dotted).

(7) Printed wall 400 (same layers as the roof), to +2.50.

(8) Buttresses 400 wide, 1000 from the inner face (600 outside), about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included); every other bay shows the printed wall (layer joints, dashed piece divisions), the rest are only outlined. Each buttress is one continuous strip 400 thick, also at the corners. Roof ribs and wall bars tie into the solid base above them.

(9*) Strip footing 1800 x 250, underside at -1.00 (hidden, dashed): printed edge formwork, cast core.

(10) Printed channel walls, 80 each side of the rib: permanent formwork, printed in the same pass as the shell (layer 2 turns into them).

(11*) Poured core: the solid base (300, full concrete over the whole 400) and the rib are poured together, in lifts of about 600 from the eave (assumed), with no joint between base and rib; dashed lines = lift joints. Printed layer joints are every 10 (drawn every 20 here).

(12*) Build-up outside the printed shell, seen at the sides of the roof: insulation 100 (type to be chosen), then the outer layer (cladding 30). The 160 air zone inside the shell stays open between the ribs.

* = assumed, to confirm

Notes

Columns stop on a horizontal rib (at 3626 and 5275 above the eave); bars are lapped / hooked into it.

Horizontal ribs at 1813, 3626, 5275, plus the 300 solid base (bottom ring) and the top ring at the crown.

Dashed lines = divisions between printed pieces on the roof, in running bond (assumed about 1500 wide, 750 high). Very faint lines = printed layer joints.

All 3D printed (after the Obayashi 3dpod paper, 2023): mortar beads are printed as permanent formwork, in 10 mm layers (joints drawn every 100 on the elevation), and the structural concrete is cast into them in lifts limited by formwork pressure; the shell, wall and buttress faces are the printed skin.

The paper fills the formwork with self-compacting steel-fibre UHPC (about 180 N/mm2 compression, 8.8 tension) and needs no rebar in principle; if you use it, the Ø16 bars in the ribs can be dropped or cut back, to be confirmed with the engineer.

Rough quantities, four faces (hips counted twice, so slightly high): ribs 325 m, 4 Ø16 each bars, steel 2049 kg, cast concrete 12.5 m3.

[illegible] assumptions for the engineer to check. Elevation shows the face square on; true lengths are 1.25 x the drawn heights.

Scale 1:50 at A3 · dimensions in mm · draft 1, 2026-10-11
pyramid-roof · hand estimates, not engineered